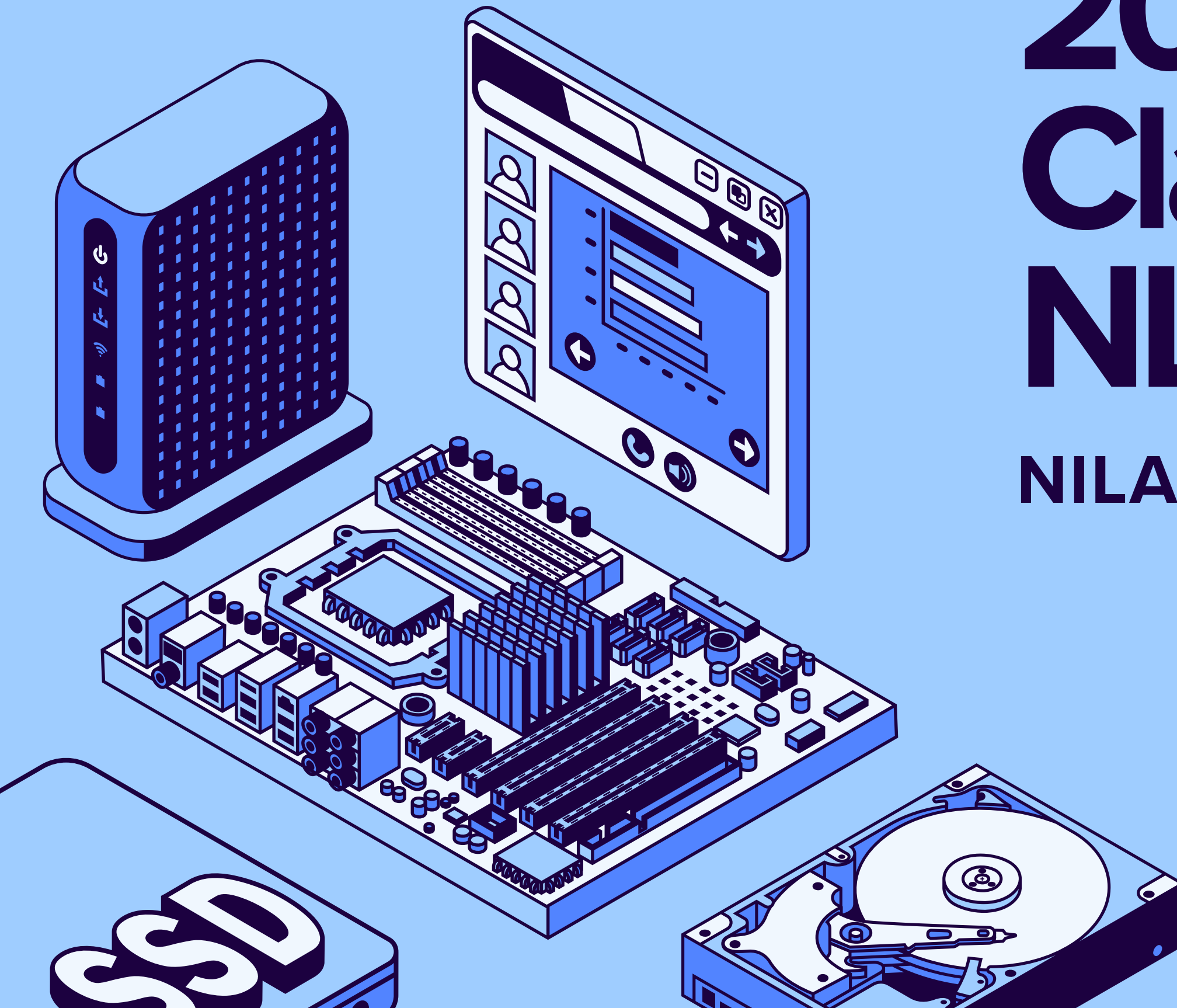


HSLU

20 Newsgroups Classification NLP Project

NILABH PANDEY

*Word2Vec / TF-IDF
BERT
MLM Domain-Adaptation
Llama-3
QLoRA*



THE PROBLEM

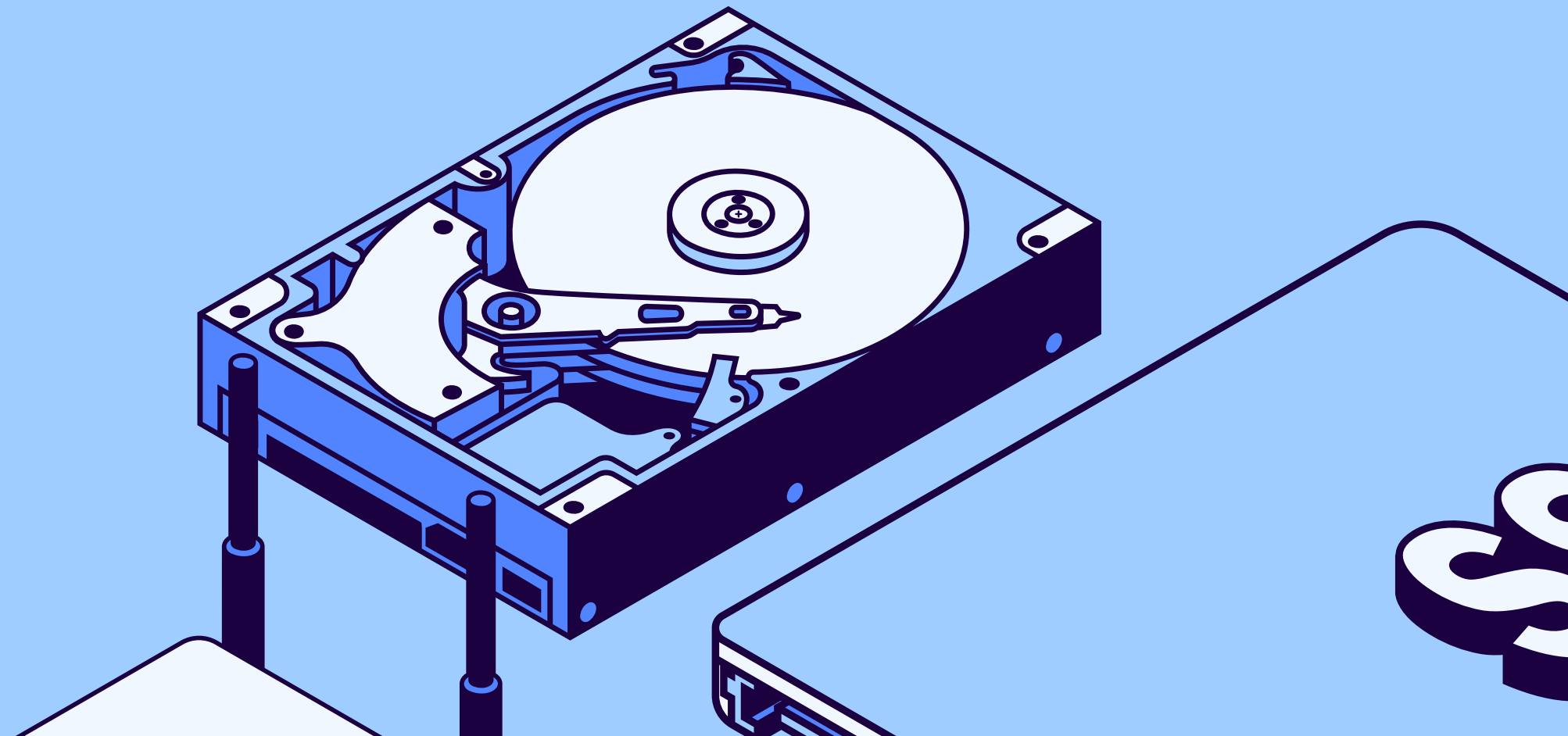
Classifying forum posts into 20 topics

Goal: automatically assign each newsgroup post to exactly one of 20 topic categories spanning religion, sports, computing, science, politics and for-sale ads.

Why it's challenging:

- 3 religion classes overlap heavily
- 4 computing classes share jargon
- 3 politics classes blur together
- short posts give few clues

20 classes · 5% baseline



20 Newsgroups corpus
18,846 documents • 20 topics

**Strip headers, footers & quoted
replies | stratified split**
9,051 train • 2,263 validation • 7,532 test

Classical pipeline
aggressive cleaning:
lowercase, stopwords,
numbers, URLs, e-mails

Task 1
Word2Vec • TF-IDF • Combined
→ small feed-forward network

67–70%

Transformer pipeline
minimal cleaning,
native subword
tokeniser

Tasks 2–4 & Bonus
BERT • BERT+MLM
Llama-3 zero/few-shot • QLoRA

47–78%

Unified evaluation
accuracy • macro-F1 • confusion matrices

THE DATASET

20 Newsgroups

18,846

Total documents

across 20 balanced classes

9,051

Training docs

stratified split

7,532

Test docs

held-out evaluation

17,886

MLM corpus

for domain adaptation

Preparation

Strip metadata

Removed headers, footers & quotes so models learn topical language, not sender artifacts

Clean & tokenize

Lowercase, remove URLs/emails/numbers, drop stopwords

Stratified split

90/10 train–val split keeps class balance across sets

METHODOLOGY

Four methods, one benchmark



Task 1

Classical features + simple NN

Word2Vec, TF-IDF, and a combined representation



Task 2

BERT fine-tuning

Full supervised transformer fine-tuning



Task 3

Domain-adaptive MLM → classify

Masked-language pretraining on in-domain text first



Task 4

Llama-3 zero-shot & few-shot

An 8B LLM with no task training

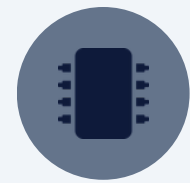


Bonus

Llama-3 QLoRA fine-tuning

Parameter-efficient fine-tuning (~0.2% of weights)

Platform migration: CPU → GPU → Pro



Phase 1

Laptop · CPU

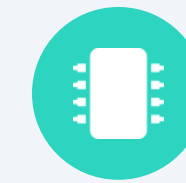
All of Task 1; BERT attempts underfit on reduced settings



Phase 2

Google Colab · T4 GPU

Proper BERT, MLM & Llama-3 training;
+7% from GPU



Phase 3

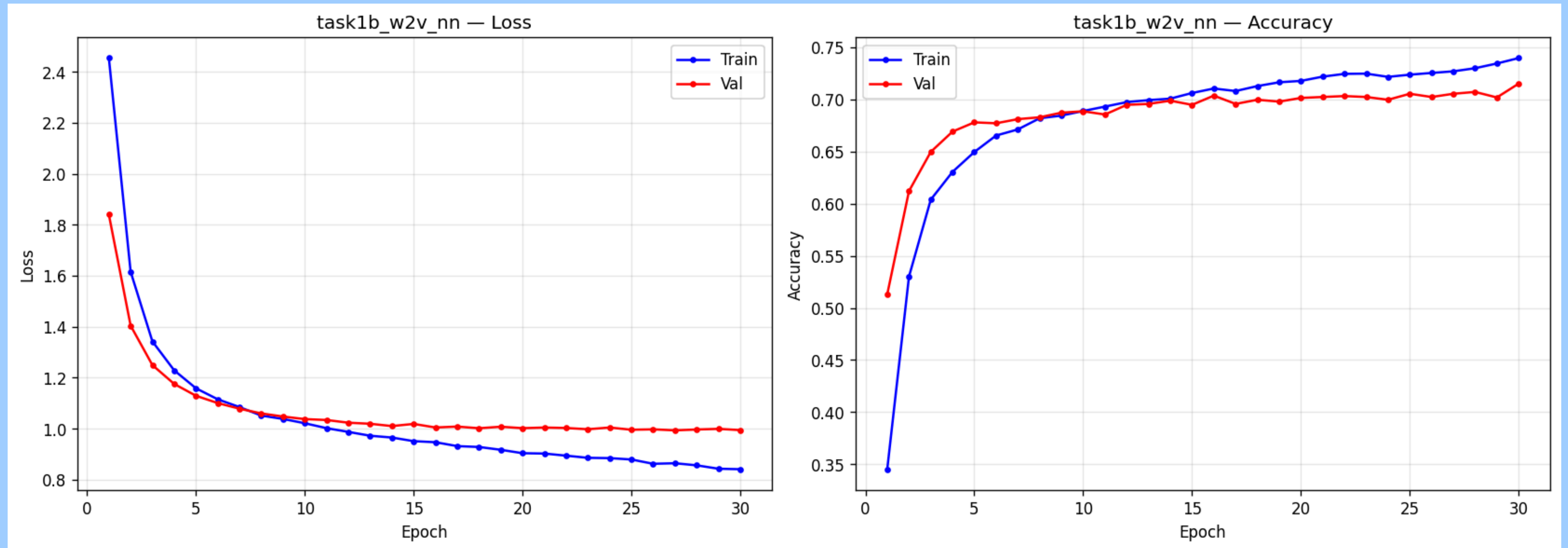
Colab Pro · L4 / A100

QLoRA bonus after hitting free-GPU limits

Tracked throughout with Weights & Biases - every run logged to one dashboard for reproducibility.
Key lesson: the same BERT model gained ~7% accuracy purely from moving CPU → GPU.

TASK 1 - CLASSICAL FEATURES

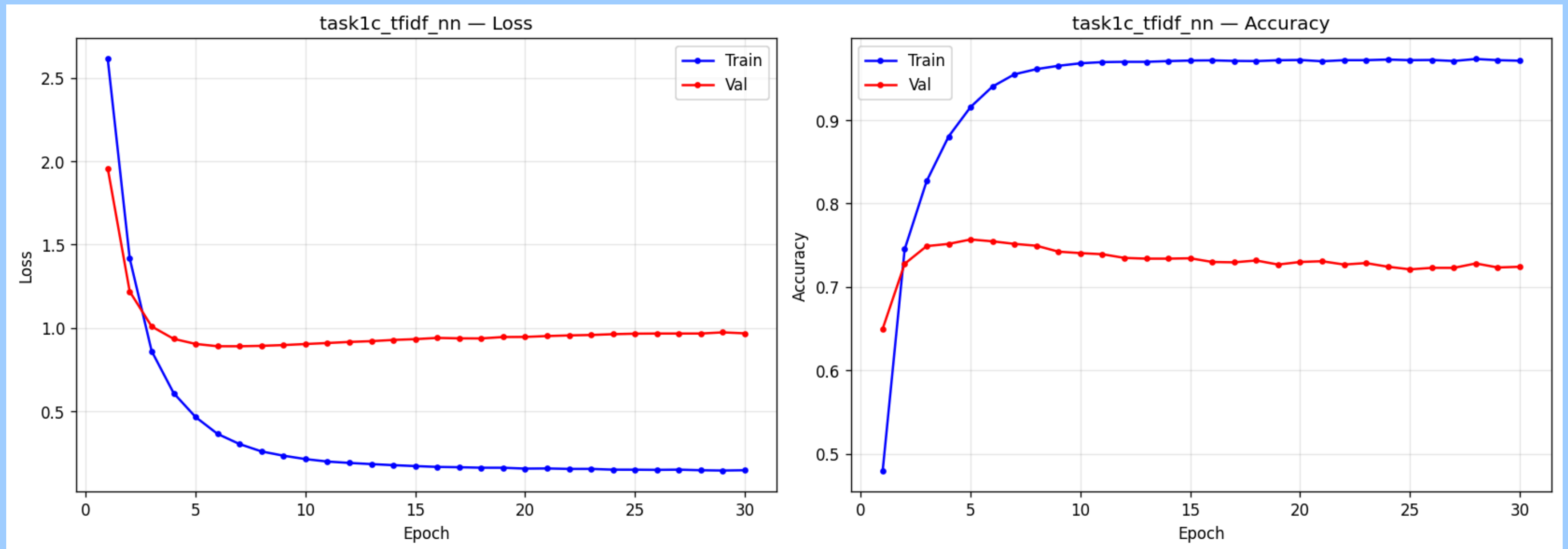
Word embeddings & TF-IDF + simple NN



W2V | Train - 73.9% | Val - 71.4%

TASK 1 - CLASSICAL FEATURES

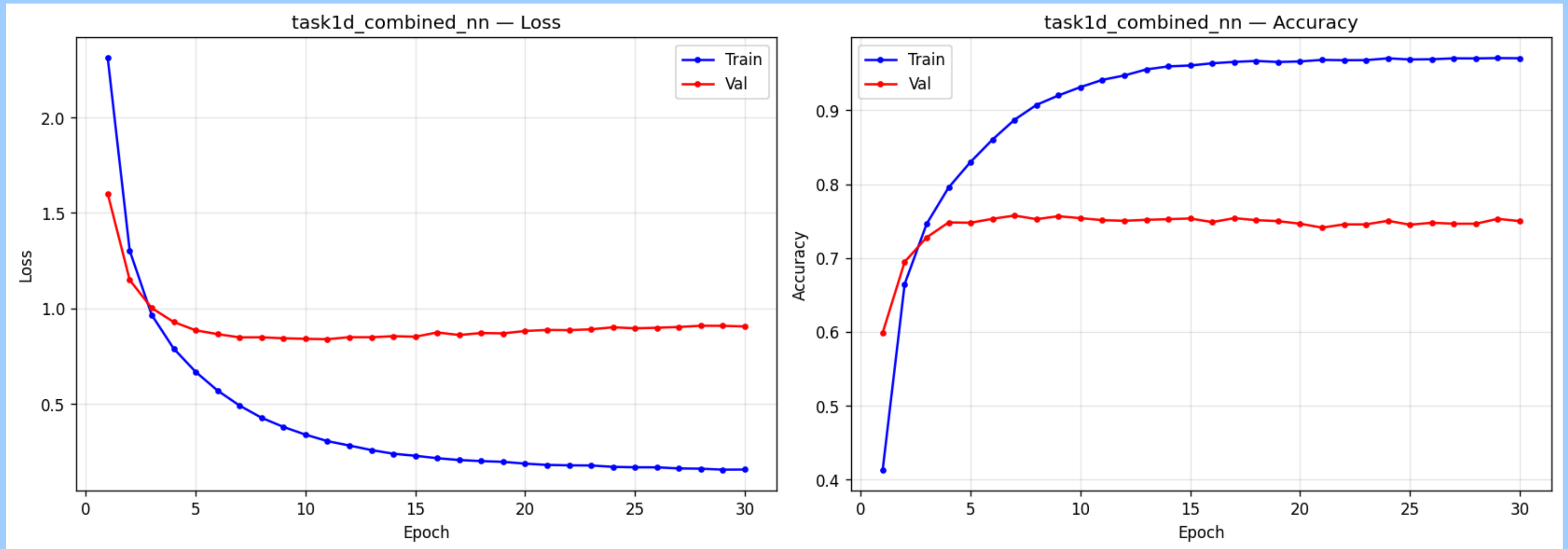
Word embeddings & TF-IDF + simple NN



TF-IDF | Train - 97.0% | Val - 72.4%

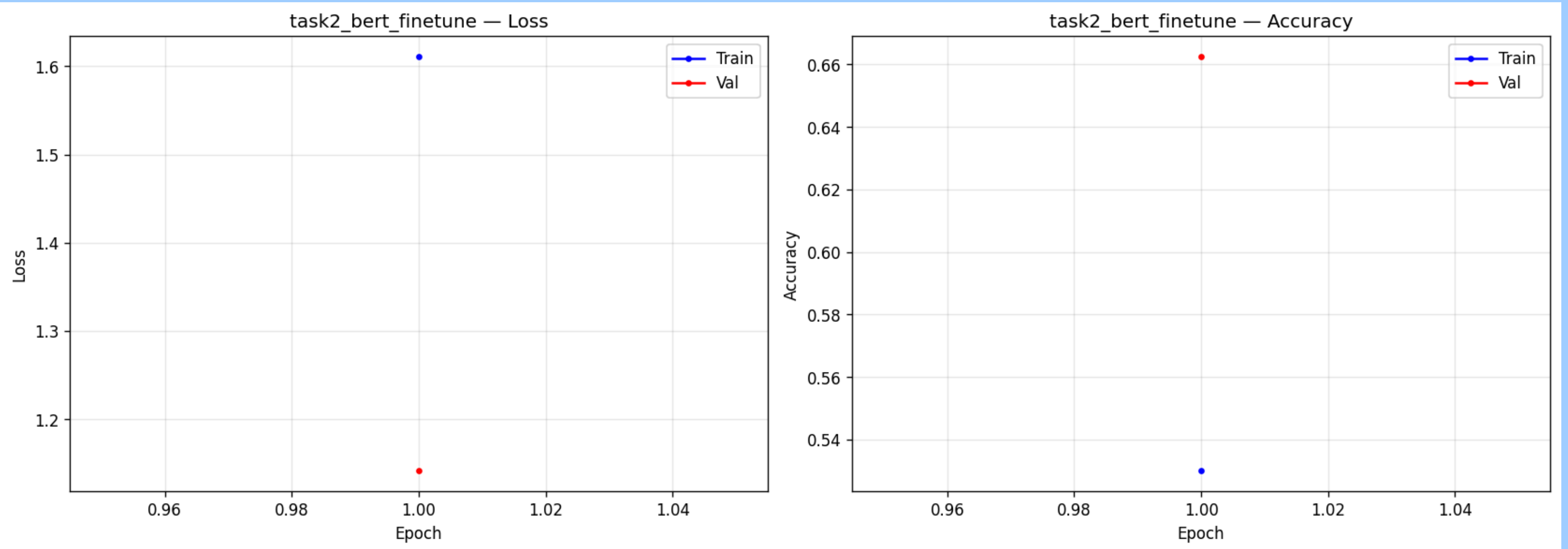
TASK 1 - CLASSICAL FEATURES

Word embeddings & TF-IDF + simple NN



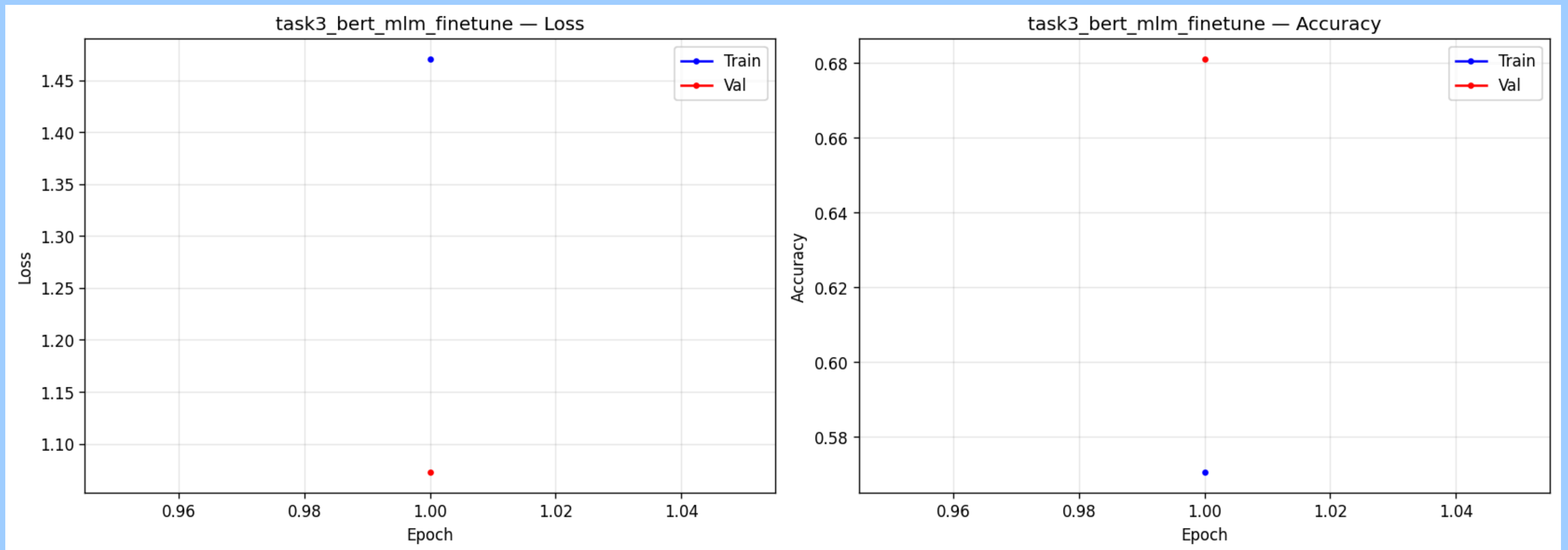
W2V + TF-IDF | Train - 97.1% | Val - 74.9%

BERT fine-tuning



Train Accuracy - 53.0% | Val Accuracy - 66.2%

Masked-language pretraining → classify



Train Accuracy - 57.0% | Val Accuracy - 68.0%

Masked-language pretraining → classify

71.5%

Test accuracy

best supervised model

11.2

MLM perplexity

down from 15.9 - adapted

17,886

Docs for MLM

in-domain, unlabeled

Two-stage recipe

Stage 1 — MLM

Mask 15% of tokens, train BERT on 17,886 in-domain docs (1 epoch)



Stage 2 — Classify

Fine-tune the adapted model on labels (max_len 256, 2 epochs)

Zero-shot & few-shot prompting

51.5%

Zero-shot

0 training examples

47.5%

Few-shot (k=3)

3 in-context examples

Surprising finding: few-shot < zero-shot

Instead of feeding Llama a plain concatenated string, we pass structured messages (a system role + a user role) through `apply_chat_template`, which wraps them in Llama-3's exact special-token format.

	v1 (raw prompt)	v2 (chat template)
Zero-shot accuracy	35.50%	51.50%
Few-shot accuracy	29.50%	47.50%

BONUS · PARAMETER-EFFICIENT FINE-TUNING

Llama-3 QLoRA fine-tuning

78.0 %

Accuracy

macro-F1 0.7465

+26.5

vs zero-shot

fine-tuning clearly helps

Configuration (mono box): Llama-3-8B-Instruct, 4-bit (nf4, double-quant, bf16) · LoRA r=16, α =32, dropout 0.05 · all 7 projection layers · 3,000 docs (150/class) · max_len 256 · completion-only loss · 2 epochs · batch 8 × accum 2 · paged_adamw_8bit · lr 2e-4 cosine · A100

WHAT'S NEXT

Future work

Stronger encoders

RoBERTa / DeBERTa and longer MLM pretraining to push past 72%

Retrieval-augmented few-shot

Pick examples similar to each post instead of random ones

Scale up QLoRA

More data + 2 epochs to close the gap to full BERT

Tackle class overlap

Targeted handling of confusable religion & politics categories

Thank You

